

KILLARA HIGH SCHOOL



YEAR 12 – 2010

HSC ASSESSMENT 3

MATHEMATICS EXTENSION 2

Time allowed: 70 minutes

Total marks: 40

- **NOT ALL** questions are of equal value.
- Start each question on a new page.
- Show all necessary working.

Question 1 (20 marks) Use a separate writing booklet

a) Evaluate $\int_0^1 \frac{e^x}{(1+e^x)^2} dx$ 2

b) By completing the square find $\int \frac{1}{\sqrt{2x-x^2}} dx$ 2

c) Find $\int \frac{8}{(x+2)(x^2+4)} dx$ 5

d) Find $\int \frac{x}{\sqrt{1-x}} dx$ 3

e) Use substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2+\cos x}$ 4

f) If $I_n = \int_1^e x(\ln x)^n dx$, $n = 0, 1, 2, 3, \dots$ 2

i. show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-2}$, $n = 1, 2, 3, \dots$

ii. Evaluate $\int_1^e x(\ln x)^3 dx$ 3

Question 2 (20 marks) Use a separate writing booklet

- a) For the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$
- Find the eccentricity. 1
 - Find the coordinates of the foci S and S' . 1
 - Find the equations of the directrices. 1
 - Sketch the curve $\frac{x^2}{16} + \frac{y^2}{25} = 1$. 2
- b) $A(a,0)$ and $A'(-a,0)$ are the vertices of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a > 0$
 $P(a \sec \theta, b \tan \theta)$ is any point on the hyperbola, $P \neq A$ or A' .
 The tangent at P meets the tangent at A, A' at Q, R respectively.
- Prove that the equation of the tangent at P is $\frac{\sec \theta}{a}x - \frac{\tan \theta}{b}y = 1$. 3
 - Find the coordinates of Q and R . 2
 - Prove that the circle with QR as a diameter passes through the two foci of the hyperbola. 5
- c) The point $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the rectangular hyperbola $xy = c^2$. The chord PQ subtends a right angle at another point $R\left(cr, \frac{c}{r}\right)$ on the hyperbola. Show that:
- $pq = -\frac{1}{r^2}$ 2
 - the normal at R is parallel to PQ . 3

Q4

$$\begin{aligned} \textcircled{a} \int_0^1 \frac{e^x}{(1+e^x)^2} dx &= \left[-\frac{1}{1+e^x} \right]_0^1 \checkmark \\ &= -\frac{1}{1+e} + \frac{1}{2} \\ &= \frac{e-1}{2(1+e)} \checkmark \end{aligned}$$

$$\begin{aligned} \textcircled{b} \int \frac{1}{\sqrt{2x-x^2}} dx &= \int \frac{1}{\sqrt{1-(x-1)^2}} dx \checkmark \\ &= \sin^{-1}(x-1) + C \checkmark \end{aligned}$$

$$\textcircled{c} \frac{a}{x+2} + \frac{bx+c}{x^2+4} = \frac{8}{(x+2)(x^2+4)} \Rightarrow a(x^2+4) + (bx+c)(x+2) = 8$$

• let $x = -2 \Rightarrow 8a = 8$
 $a = 1$

• coef. of constant terms : $4a + 2c = 8$
 $2c = 4 \Rightarrow c = 2 \checkmark$

• coef of x^2 terms : $a + b = 0 \Rightarrow b = -1$

$$\begin{aligned} \therefore \int \frac{8}{(x+2)(x^2+4)} dx &= \int \frac{1}{x+2} + \frac{2}{x^2+4} - \frac{x}{x^2+4} dx \checkmark \\ &= \ln(x+2) + \sqrt{\tan^{-1} \frac{x}{2}} - \frac{1}{2} \ln(x^2+4) + C \checkmark \end{aligned}$$

$\textcircled{d}$ let $u = \sqrt{1-x}$ $\Rightarrow du = \frac{-dx}{2\sqrt{1-x}} \checkmark$
 and $x = 1-u^2$

$$\begin{aligned} \therefore \int \frac{x}{\sqrt{1-x}} dx &= -2 \int 1-u^2 du \\ &= -2 \left(u - \frac{1}{3} u^3 \right) \checkmark \\ &= \frac{2}{3} (1-x)^{3/2} - 2\sqrt{1-x} + C \checkmark \end{aligned}$$

$$\textcircled{e} \quad \text{let } t = \tan \frac{x}{2} \Rightarrow dt = \frac{1}{2} \sec^2 \frac{x}{2} dx \\ = \frac{1}{2} (1+t^2) dx \Rightarrow dx = \frac{2dt}{1+t^2}$$

$$\text{when } x = 0, t = 0 \\ x = \frac{\pi}{2}, t = \tan \frac{\pi}{4} = 1$$

$$\therefore \int_0^{\pi/2} \frac{dx}{2+\cos x} = \int_0^1 \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\ = 2 \int_0^1 \frac{1}{2+2t^2+1-t^2} dt \\ = 2 \int_0^1 \frac{1}{t^2+3} dt \\ = \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 \\ = \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6} = \frac{\pi\sqrt{3}}{9}$$

$$\textcircled{f} \quad I_n = \int_1^e x (\ln x)^n dx, \quad n=0, 1, 2, \dots \\ = \left[\frac{1}{2} x^2 (\ln x)^n \right]_1^e - \frac{1}{2} \int_1^e x^2 \cdot n \cdot \frac{1}{x} (\ln x)^{n-1} dx \\ = \frac{e^2}{2} - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx \\ \therefore I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1} \quad n=1, 2, 3, \dots$$

$$\text{(ii)} \quad I_3 = \int_1^e x (\ln x)^3 dx \\ = \frac{e^2}{2} - \frac{3}{2} \left(\frac{e^2}{2} - 1 \cdot \left(\frac{e^2}{2} - \frac{1}{2} I_0 \right) \right) \quad \text{where } I_0 = \int_1^e x dx \\ = \left[\frac{x^2}{2} \right]_1^e \\ = \frac{e^2}{2} - \frac{1}{2} \\ \therefore I_3 = \frac{e^2}{2} - \frac{3e^2}{4} + \frac{3e^2}{4} - \frac{3}{4} \cdot \frac{e^2}{2} + \frac{3}{8} \\ = \frac{e^2}{8} + \frac{3}{8} = \frac{1}{8} (e^2 + 3)$$

(Q2)

(4)

$$(a) \frac{x^2}{16} + \frac{y^2}{25} = 1$$

$$(i) 16 = 25(1 - e^2)$$

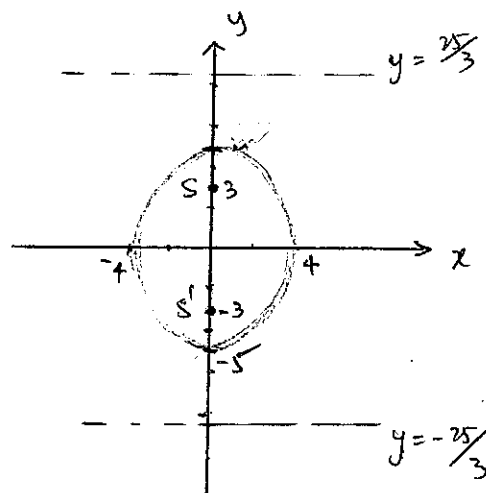
$$e^2 = 1 - \frac{16}{25}$$

$$= \frac{9}{25}$$

$$e = \frac{3}{5} \checkmark$$

$$(ii) S(0, \pm 5 \times \frac{3}{5}) \Rightarrow S(0, \pm 3) \checkmark$$

$$(iii) y = \pm \frac{b}{e} \Rightarrow y = \pm 5 \div \frac{3}{5} \Rightarrow y = \pm \frac{25}{3} \checkmark$$



(b)

gradient of the tangent at P

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{bx}{ay}$$

$$\text{at } \begin{cases} x = a \sec \theta \\ y = b \tan \theta \end{cases}$$

$$\therefore m = \frac{b \sec \theta}{a \tan \theta} \checkmark$$

equation of the tangent at P

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta) \checkmark$$

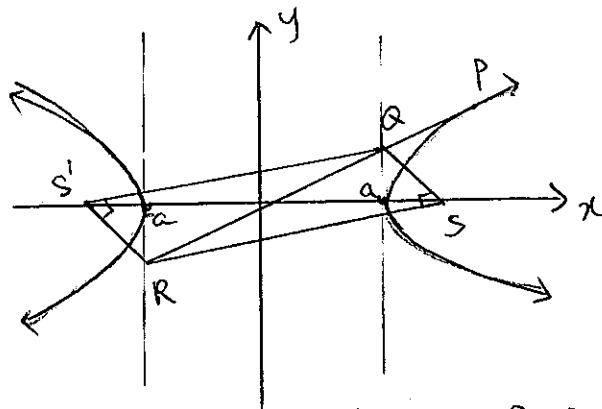
$$\frac{y}{b} \tan \theta - \frac{b \tan^2 \theta}{b} = \frac{x}{a} \sec \theta - \frac{b \sec^2 \theta}{a}$$

$$\Rightarrow \frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = \tan^2 \theta - \sec^2 \theta$$

$$\frac{\sec \theta}{a} x - \frac{\tan \theta}{b} y = 1 \checkmark$$

(ii)

(5)



from (i) equation of the tangent at P is $\frac{\sec \theta}{a} x - \frac{\tan \theta}{b} y = 1$ (1)

Sub. $x=a$ into (1) $\Rightarrow \frac{a}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$

$$\Rightarrow y = (\sec \theta - 1) \frac{b}{\tan \theta}$$

$\therefore$ Coordinates of Q is $(a, \frac{b}{\tan \theta} (\sec \theta - 1))$ ✓

Sub. $x=-a$ into (1) $\Rightarrow -\sec \theta - \frac{y}{b} \tan \theta = 1$

$$\Rightarrow y = -\frac{b}{\tan \theta} (\sec \theta + 1)$$

$\therefore$ Coordinates of R is $(-a, -\frac{b}{\tan \theta} (\sec \theta + 1))$ ✓

Now $m_{QS} = \frac{b}{\tan \theta} (\sec \theta - 1) \div (a - ae)$

and $m_{RS} = \frac{b}{\tan \theta} (\sec \theta + 1) \div (a + ea)$

$$m_{QS} \cdot m_{RS} = \frac{b^2}{\tan^2 \theta} (\sec^2 \theta - 1) \div (a^2 - a^2 e^2) \quad \checkmark$$

$$= b^2 \div -a^2 (e^2 - 1)$$

$$= b^2 \div -b^2$$

$$= -1$$

$\therefore \angle QRS = 90^\circ$ (QS $\perp$ RS). ✓

similarly

$$m_{RS'} = \frac{-b}{\tan \theta} (\sec \theta + 1) \div (-a + ae)$$

$$m_{QS'} = \frac{b}{\tan \theta} (\sec \theta - 1) \div (a + ae)$$

$$m_{RS'} \cdot m_{QS'} = \frac{b^2}{\tan^2 \theta} (\sec \theta - 1) \div (a^2 - a^2 e^2)$$

$$= b^2 \div [-a^2(e^2 - 1)]$$

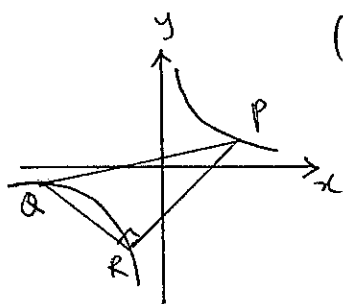
$$= -1$$

$$\therefore RS' \perp QS'$$

both $\angle QS'R$ and $\angle QSR$ are angles in a semi circle with diameter QR

$\therefore S, S'$ lie on the circle with diameter QR .

(c)



$$(i) \quad m_{PR} = \frac{\frac{c}{P} - \frac{c}{R}}{cp - cr} = \frac{c(R-P)}{cPR(P-R)} = -\frac{1}{pr}$$

Similarly

$$m_{QR} = -\frac{1}{qr}$$

$$\text{Since } QR \perp PR, \quad m_{QR} \times m_{PR} = -1$$

$$\Rightarrow -\frac{1}{pr} \cdot -\frac{1}{qr} = -1 \Rightarrow \frac{1}{pqr^2} = -1$$

$$\therefore pq = -\frac{1}{r^2}$$

$$(ii) \text{ Now } m_{PQ} = -\frac{1}{pq} \text{ sub. } pq = -\frac{1}{r^2}$$

$$\Rightarrow m_{PQ} = -\frac{1}{-\frac{1}{r^2}} = r^2$$

Gradient of the tangent at P

$$x \frac{dy}{dx} + y = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

$$\text{at } x = cr, \quad y = \frac{c}{r} \Rightarrow \frac{dy}{dx} = -\frac{c}{r} \div cr$$

$$= -\frac{1}{r^2}$$

$\therefore$ gradient of the normal at R is r^2

$$m_{PQ} = m_R = r^2$$

$\therefore$ PQ is // the tangent at R. ✓